Reindexing changes the labels of a basis. Applying e and then f uses the composite f after e; evaluating the resulting basis follows the inverse maps.

Result 1 (successive basis reindexing). [Basis transport, composition law from Module.Basis.reindex_apply]
Let K, I, J and L be types. Let *field* be a field structure on K. Let M be a module over K. Let b be a basis of M indexed by I. Let e be an equivalence from I to J. Let f be an equivalence from J to L. Then

$$b[e][f] = b[(f \circ e)].$$

Adapted from the cited source.

Result 2 (identity basis reindexing). [Basis transport, identity law from Module.Basis.reindex_apply]
Let K and I be types. Let *field* be a field structure on K. Let M be a module over K. Let b be a basis of M indexed by I. Then

$$b[\mathrm{id}_I] = b.$$

Adapted from the cited source.

Result 3 (reversing basis reindexing). [Basis transport, inverse law from Module.Basis.reindex_apply]
Let K, I and J be types. Let *field* be a field structure on K. Let M be a module over K. Let b be a basis of M indexed by I. Let e be an equivalence from I to J. Then

$$b[e][e^{-1}] = b.$$

Adapted from the cited source.

LLM-assisted Clean statements and Lean proofs (gpt-6, 2026-11-11). See PROVENANCE.md. Composition applies the first equivalence, then the second; basis evaluation uses the inverse direction. These laws do not prove a basis-extension theorem.